

have a single-channel, 64-bit memory controller, whereas the Socket 939 has a 128-bit, dual-channel controller. The HyperTransport link—a serial bus linking components on an AMD motherboard—on a 754-pin processor is capped at 1600 MHz, which is otherwise 2000 MHz for the 939-pin processor. The total processor-to-system bandwidth on a 754-pin is limited to 9.6 GB/s, whereas it is 14.4 GB/s for 939s. In short, the 939-pin processors are geared for better performance than the 754s.

The L2 cache size on the Athlon 64 processor varies between 512 KB and 1 MB. Similar to Intel's 'EE' processors, AMD has the Athlon 64 FX range, available only on a 939-pin package.

Sempron processors are meant to replace the older Athlon XP range of processors. They are available in two versions—one for the 754 Socket and the other for the soon-to-retire Socket 'A'. Why two versions?



A 754-pin Sempron 3100+ and a Socket 'A' Sempron 2800+

Initially, AMD wanted to retire the Athlon XP range, but since the Socket 'A' platform provides the most value, they thought it wise to postpone its death. Hence the release of Sempron for Socket 'A' motherboards. Moreover, to bring some stability to their platforms, AMD identifies the Socket 754 as the ideal platform for the near future, and hence the release of the 754 package Sempron.

AMD processors of yesteryear were infamous for their heating problems, but this is no longer the case.

The Athlon 64 hardly gets past 50 degrees Celsius; in et, suggesting a CPU-memory link inferior to the e Gigabyte board.

Half-Life 2 scales well with processor performance. Since MSI's

915P Neo 2 did well on the PCMark CPU score, it came as no surprise when the board offered the the board offered system hangs and crashes are experienced.

Who's The King?

Let's get into the real comparison of the processors and see who comes out tops. This comparison included six processors from Intel, ranging from clock speeds of 2.53 GHz to 3.6 GHz.

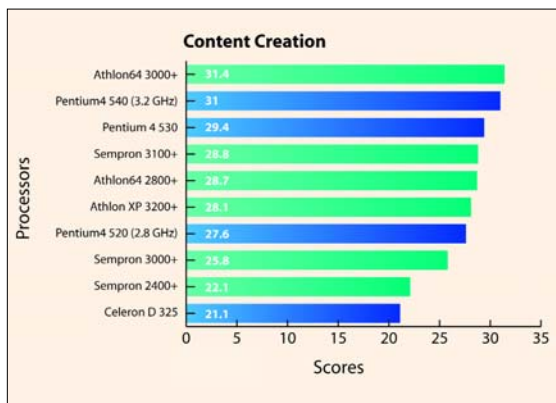
AMD provided us with 10 processors, including the top-of-the-line FX-55, along with the entire Sempron range. We also included the Athlon XP 3200+. It should be noted that high-end processors such as the P4 3.4 GHz EE, Athlon FX-55, Athlon FX-53, and the Athlon XP 3200+ are either too costly or are phased out, and hence are not part of this comparison. We included them to bring better clarity to the results of the mid-range processors. Let us take each benchmark and dissect it to see which processor fares well and why.

Content Creation 2004

This is a system benchmark, and so, apart from raw processing power, it also relies on other critical parameters such as I/O bandwidth and memory bandwidth. The Athlon64 3000+ was right on top, scoring 31.4, despite the 1.2 GHz advantage the Pentium 4 3.2 GHz Intel processor has.

The Athlon64 3000+ processor has an on-die, dual-memory controller, and hence, memory bandwidth is not an issue and stands at 8 GB/s. Intel's 3.6 GHz also supports dual-channel, but its memory controller is part of the Northbridge and provides a bandwidth of 6.4 GB/s.

When you look at the results, the difference in performance, though not significant at first



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